

Joanna X. Ju

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LinkedIn – www.linkedin.com/in/joannaxiaolinju

Portfolio (GitHub) – <https://github.com/joannaxju>

Summary

Computer Engineering student at UC San Diego with a strong interest in processor design and digital hardware. Experience designing a custom 8-bit RISC processor and developing embedded systems and firmware through the UCSD Rocket Propulsion Laboratory Avionics Team. Skilled in SystemVerilog, RTL design, FPGA development, C/C++, and low-level programming, with a focus on developing more complex, performance-oriented processor architectures.

Education

University of California, San Diego – La Jolla, CA

Fall 2025 – Present

Bachelor of Science in Computer Engineering

Expected Graduation: Spring 2028

- Cumulative GPA: 3.70 / 4.00 (Transferred from Pasadena City College, Pasadena, CA)

Projects

Custom 8-bit RISC Processor Design – SystemVerilog, Assembly, C++

- Designed and implemented a custom 8-bit RISC processor, defining the ISA, instruction formats, registers, and addressing modes
- Implemented the processor datapath, ALU, control logic, and memory interfaces using synthesizable SystemVerilog
- Developed assembly programs and an assembler to translate assembly instructions into machine code and resolve label offsets
- Verified processor functionality through simulation and FPGA testing, using results to refine the ISA and implementation

Flight Computer Software Simulator (RPL Software) – C++, Arduino, Ethernet (*In Progress*)

- Developing a simulation environment for an Arduino-based flight computer and ground station, modeling sensor inputs, actuator control, and system states

Experience

SPIS Mentor – UCSD CSE Department, La Jolla, CA

August 2026 – Present

- Mentor a cohort of incoming computer science and engineering students throughout the SPIS program
- Guide students in designing, programming, and debugging Raspberry Pi-based robotics projects using Python
- Grade algorithmic problem-solving homework and programming labs, providing constructive feedback to reinforce core computer science concepts

Avionics Team Member – UCSD Rocket Propulsion Laboratory (RPL), La Jolla, CA

October 2025 – Present

- Collaborate with the avionics team to maintain and extend embedded systems for student-built rockets
- Analyze C++ and Arduino-based firmware to understand system architecture, hardware integration, and sensor communication
- Develop and test new firmware functions to ensure reliable embedded system performance
- Prototype and integrate circuits connecting sensors and electronic components to microcontrollers

Math Instructor – Mathnasium, Temple City & San Diego, CA

July 2024 – Present

Computer Science Tutor – Pasadena City College, Pasadena, CA

August 2024 – June 2025

Skills

Processor Design & Digital Systems: Computer Architecture, Processor Design, Digital Design, RTL Design, FPGA, SystemVerilog, Verilog, Datapath Design, Control Logic

Embedded Systems: Embedded Systems, Microcontrollers, Arduino, Sensor Integration, Hardware-Software Integration

Programming: C/C++, Python, Java, Assembly

Hardware & Electrical Engineering: Analog & Digital Circuit Design, Cadence Virtuoso, Circuit Simulation, PSpice, LTspice, Oscilloscope, Multimeter

Tools: Git, GitHub, Quartus, ModelSim/Quarta, Arduino IDE, VS Code

Languages: English (Fluent), Mandarin Chinese (Fluent)